Physics - Undergraduate Level Assessment 8 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The emission spectrum of the doubly ionized lithium atom Li++ (Z = 3, A = 7) is identical to that of a hydrogen atom in which all the wavelengths are
   (a) decreased by a factor of 9
   (b) decreased by a factor of 49
   (c) decreased by a factor of 81
   (d) increased by a factor of 9

2. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
   (a) 0.1 GeV/c^2
   (b) 0.2 GeV/c^2
   (c) 0.5 GeV/c^2
   (d) 1.0 GeV/c^2

3. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
   (a) Diode laser
   (b) Dye laser
   (c) Free-electron laser
   (d) Gas laser

4. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
   (a) Constant temperature
   (b) Constant volume
   (c) Constant pressure
   (d) Adiabatic

5. Which of the following gives the total spin quantum number of the electrons in the ground state of neutral nitrogen (Z = 7)?
   (a) 1/2
   (b) 1
   (c) 3/2
   (d) 5/2

True / False (10 marks)

Answer True or False

6. True or False: If the total energy of a particle is twice its rest energy, its relativistic momentum is ($\sqrt{}$ 3)mc.
7. True or False: The sign of the charge carriers in a doped semiconductor can be deduced from its specific heat measurement.
8. True or False: In a constant pressure process, the increase in the internal energy of an ideal gas is equal to the heat added to the gas.
9. True or False: In the observer's reference frame, it takes 2.5 ns for the meter stick traveling at 0.8 c to pass the observer.
10. True or False: A reversible thermodynamic process always results in an increase in the entropy of the system and its environment.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the work required to accelerate a particle of mass m from rest to a speed of 0.6 c. Discuss the implications of relativistic effects in your analysis.
12. Discuss the measurements an astronomer can derive from observing a moon's orbital dynamics around a planet, and explain why the moon's mass cannot be calculated from these measurements.

End of Paper